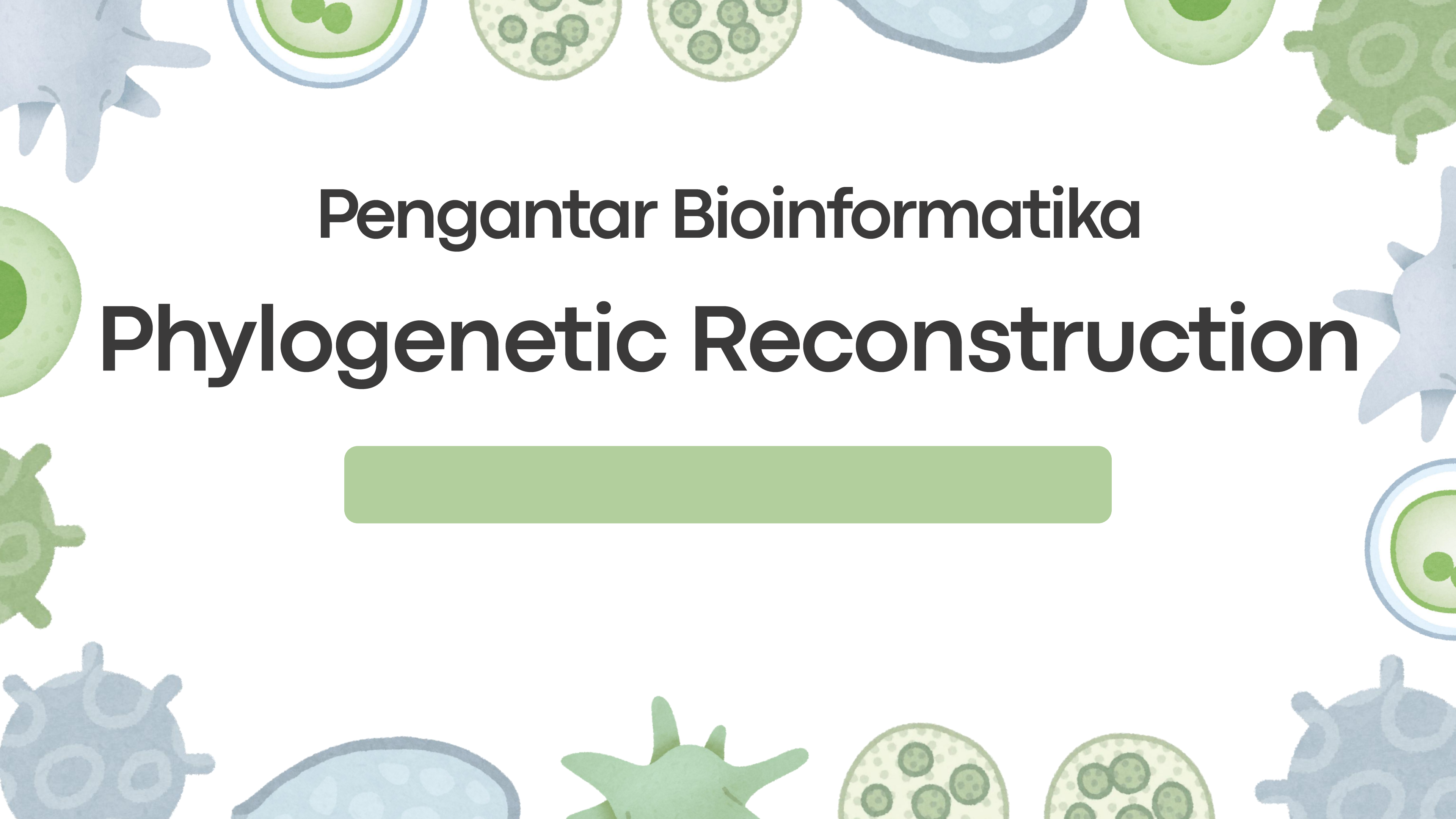




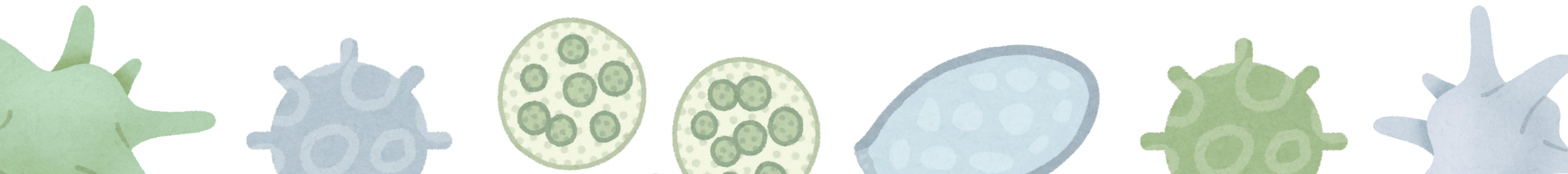
Pengantar Bioinformatika

Phylogenetic Reconstruction



Phylogenetic Trees

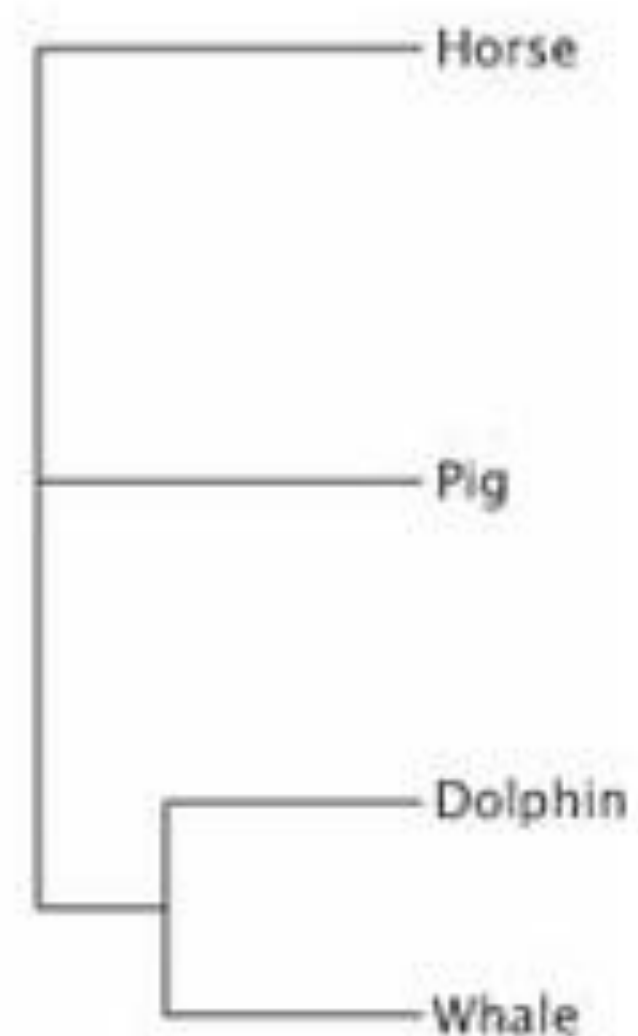
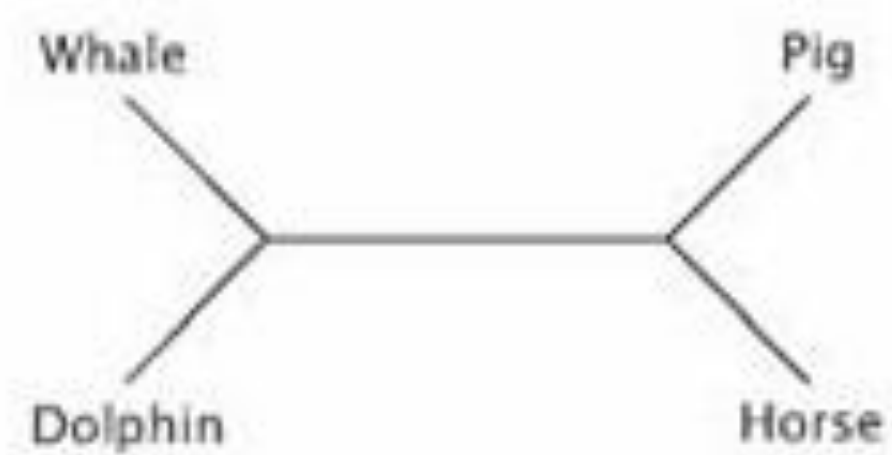
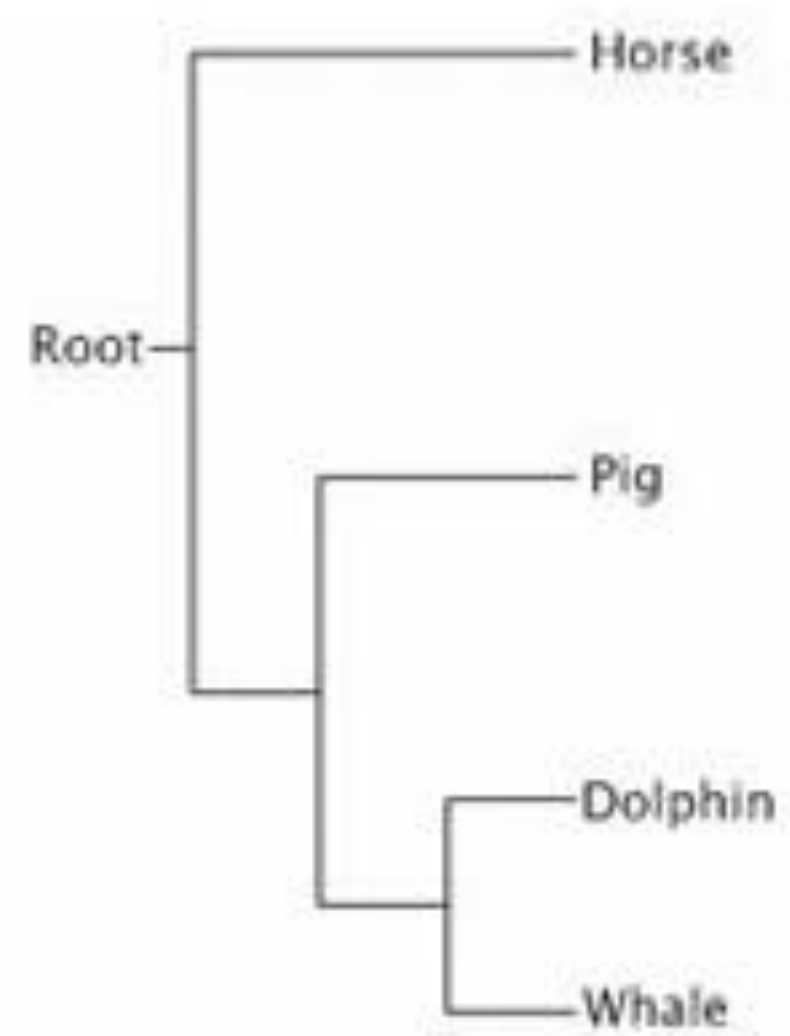
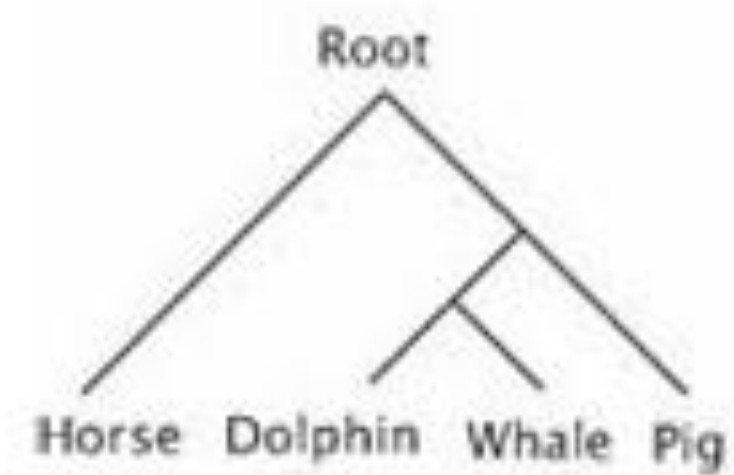
- Phylogenetic trees graphically represent evolutionary relationships among species or genes.
- Operational Taxonomic Units (OTUs) are the species or sequences studied, labeling the tree's leaves.
- A rooted tree descends from a single root, representing the common ancestor and showing evolutionary direction.
- Internal nodes in a rooted tree represent ancestors of specific OTU groups.
- An unrooted tree specifies only evolutionary relationships, lacking a root or evolutionary direction.
- Tree topology is the branching pattern, distinct from branch lengths.
- Branch length is a positive number indicating relatedness, often calculated as $(\text{time} \times \text{evolutionary rate})$.
- A key assumption is that divergence occurred "instantaneously" relative to branch lengths.

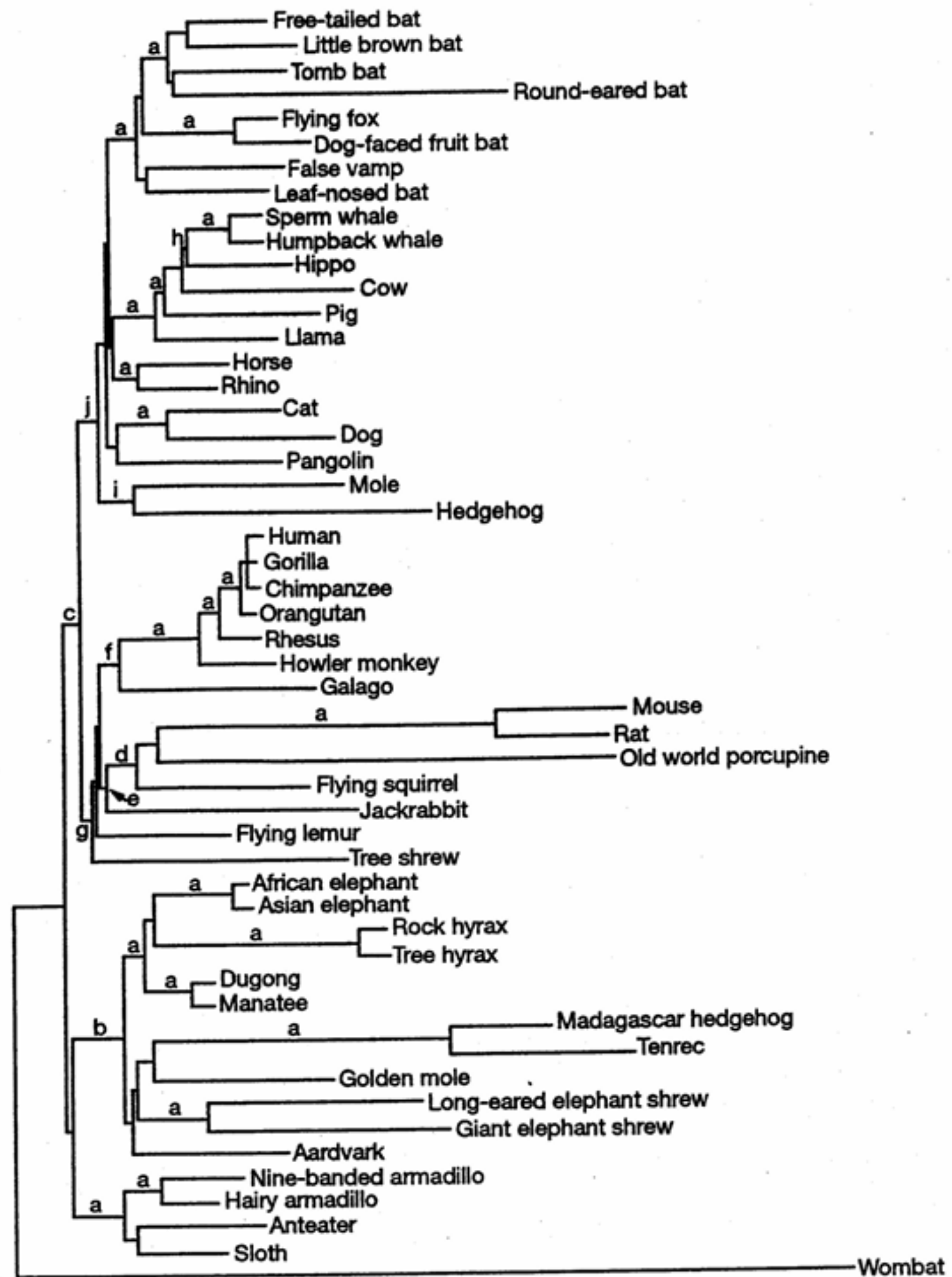


1. Pendahuluan: Phylogenetic Trees (Pohon Filogenetik)

1.1 Definisi Dasar

- **Phylogenetic tree:** Diagram yang merepresentasikan hubungan evolusioner antar spesies atau gen.
- **Rooted tree:** Pohon yang memiliki akar (root) sebagai nenek moyang bersama, bercabang menjadi dua di setiap node internal.
- **Unrooted tree:** Hanya menunjukkan hubungan antar OTU (Operational Taxonomic Units) tanpa arah evolusi.
- **Branch length:** Panjang cabang yang merepresentasikan derajat kekerabatan (biasanya hasil kali waktu dan laju evolusi).
- **Tree topology:** Pola percabangan tanpa panjang cabang.





1.2 Jumlah Topologi Pohon

Untuk N OTU:

- Rooted trees:

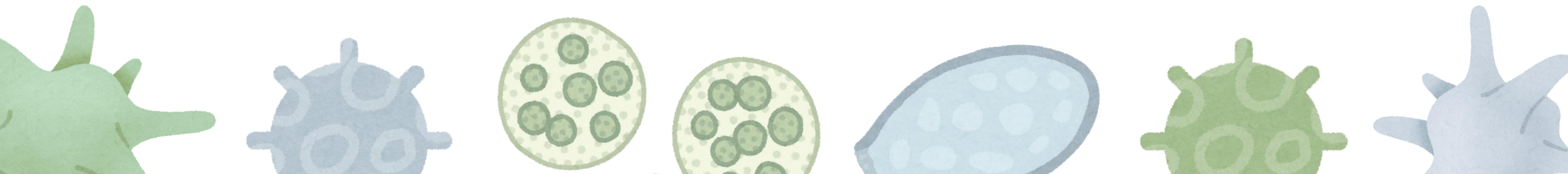
$$\frac{(2N - 3)!}{2^{N-2}(N - 2)!}$$

- Unrooted trees:

$$\frac{(2N - 5)!}{2^{N-3}(N - 3)!}$$

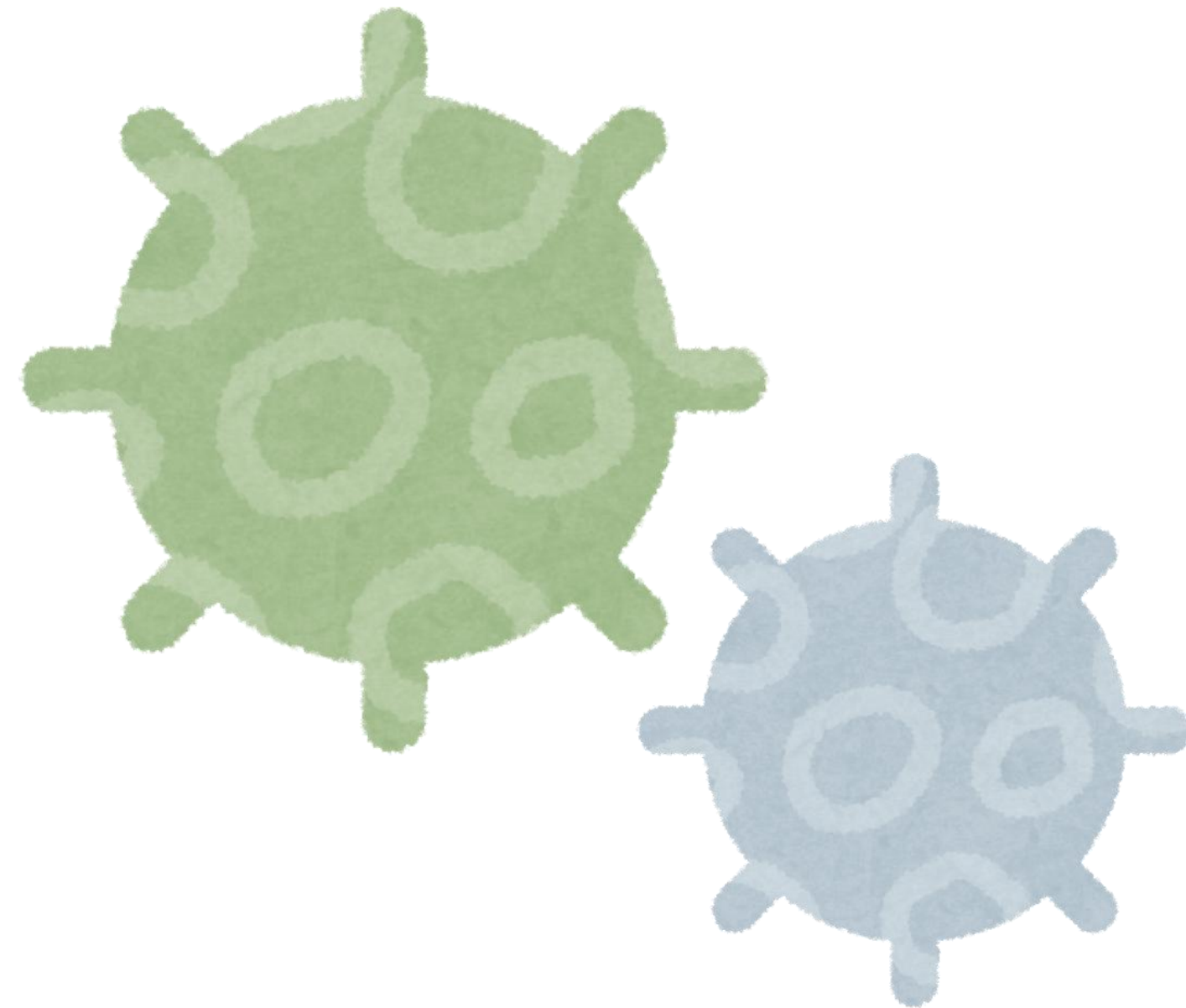
Phylogenetic Trees

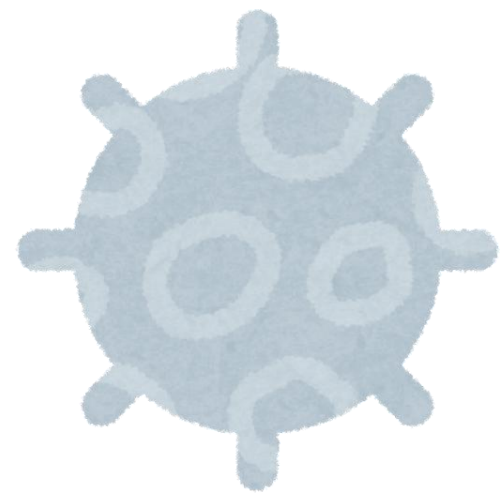
- The number of unrooted topologies for N OTUs is $\frac{(2N-5)!}{2^{N-2}(N-2)!}$.
- The number of rooted topologies for N OTUs is $\frac{(2N-3)!}{2^{N-2}(N-2)!}$.
- Modern methods use sensitive gene/protein sequences, as sequence divergence precedes morphological divergence.
- It is important to differentiate between gene trees and species trees.
- Orthologues are genes diverged by speciation, paralogues are genes diverged by duplication.
- Phylogenetic reconstruction is the process of building an optimal tree from sequence data.
- The process involves (i) choosing homologous OTUs, (ii) creating a reduced multiple alignment (removing gaps), and (iii) inferring the tree.
- Main inference methods include (a) Parsimony, (b) Distance methods, and (c) Probabilistic methods (e.g., Maximum Likelihood).



Parsimony Methods

- Parsimony finds rooted topologies and ancestral sequences, not branch lengths.
- The main goal is finding the "most parsimonious topology" (lowest total cost).
- It assumes evolution minimizes cost (e.g., total substitutions).
- A topology's cost is the minimum sum of substitutions across all branches.
- Fitch's Algorithm quickly calculates the cost for one fixed topology.
- Searching all topologies is computationally difficult for large N (number of OTUs).
- The Branch and Bound Algorithm (with Fitch's) efficiently finds all parsimonious trees.
- Parsimony is the fastest reconstruction method but is considered among the least accurate.





An Example of Parsimony Methods

Example 5.1. Let $\mathcal{Q}$ be the three-letter alphabet $\{A, B, C\}$, and suppose that we are given the following reduced multiple alignment of three sequences

$x^1 : A A B$

$x^2 : A C B$

$x^3 : C C B.$

There are only three possible rooted topologies for three OTUs, and it is easy to observe that each topology is most parsimonious for x^1, x^2, x^3 . Figure 5.4 shows the topologies together with all optimal sequence assignments. The cost of each topology is equal to 2.

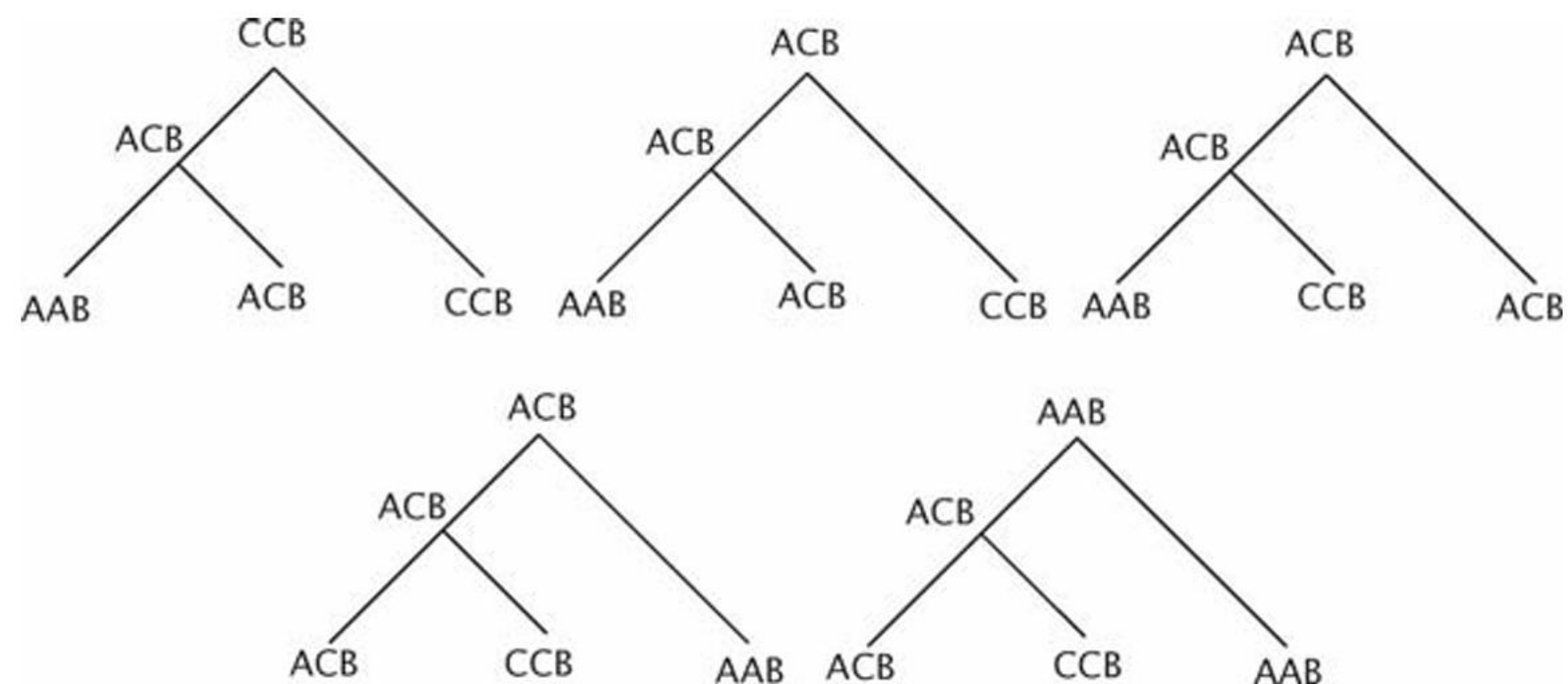


Fig. 5.4.

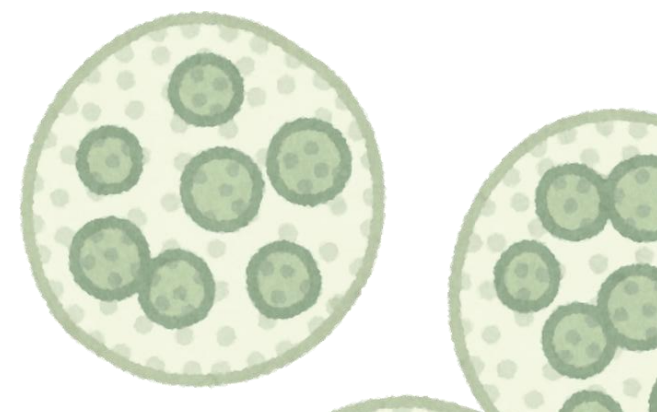
Distance Methods - Foundations

- Distance methods reconstruct trees (rooted or unrooted) from a matrix of pairwise distances between sequences.
- A function d is a distance function if it satisfies:
 - (i) $d(u, v) > 0$ (if $u \neq v$)
 - (ii) $d(u, u) = 0$
 - (iii) $d(u, v) = d(v, u)$ (Symmetry)
 - (iv) $d(u, v) \leq d(u, w) + d(w, v)$ (Triangle Inequality)
- These distances are represented by a symmetric distance matrix $M_d = (d_{ij})$.
- The goal is to find a tree T where the tree-generated distance d_T (path length between OTUs) is as close as possible to the given d .
- A distance function d is additive if a tree T exists such that $d_T = d$ (the tree perfectly represents the distances).
- For $N = 3$ OTUs, any distance function is additive. The branch lengths x, y, z are found using:

$$x = \frac{1}{2}(d_{12} + d_{13} - d_{23})$$

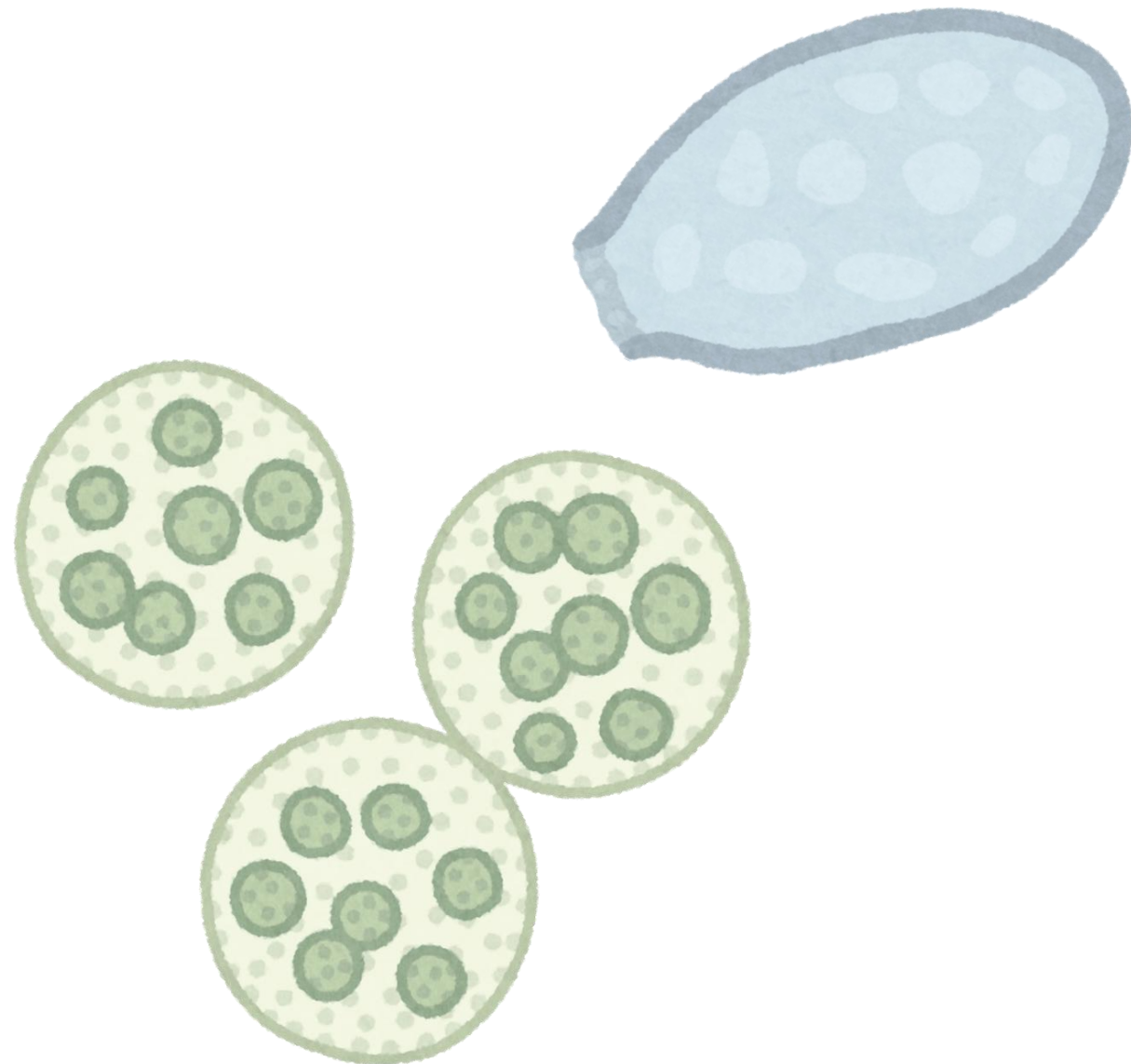
$$y = \frac{1}{2}(d_{12} + d_{23} - d_{13})$$

$$z = \frac{1}{2}(d_{13} + d_{23} - d_{12})$$



Distance Methods - The Neighbor-Joining (NJ) Algorithm

- For $N \geq 4$, not all distance functions are additive.
- **Theorem 5.3 (Four-Point Condition):** A distance d is additive if and only if for any four OTUs i, j, k, l , two of the sums $(d_{ij} + d_{kl}, d_{ik} + d_{jl}, d_{il} + d_{jk})$ are equal and are greater than or equal to the third.
- The **Neighbor-Joining (NJ) Algorithm** is a method that constructs the unique tree for any additive distance function.



- **NJ Algorithm Steps:**

1. Calculate r_i for each OTU (average distance): $r_i = \frac{1}{N-2} \sum_{k=1}^N d_{ik}$
 2. Calculate D_{ij} matrix entries: $D_{ij} = d_{ij} - (r_i + r_j)$
 3. Find the pair (i, j) with the **minimal** D_{ij} . This pair is "joined" by a new node, x_{N+1} .
 4. Calculate new branch lengths to this node:
 - $d_{N+1,i} = \frac{1}{2}(d_{ij} + r_i - r_j)$
 - $d_{N+1,j} = \frac{1}{2}(d_{ij} + r_j - r_i)$
 5. Calculate distances from the new node to all others m :
 - $d_{N+1,m} = \frac{1}{2}(d_{im} + d_{jm} - d_{ij})$
 6. Repeat the process with $N - 1$ nodes until only 3 remain.
- **Pseudodistance:** Real data often violates the four-point or triangle conditions. Applying NJ to such "pseudodistances" can cause anomalies, like negative branch lengths.

An Example of The Neighbor-Joining (NJ) Algorithm

Example 5.4. Let $N = 6$ and suppose that we are given the following distance matrix

$$\begin{array}{c|cccccc} M_d & x^1 & x^2 & x^3 & x^4 & x^5 & x^6 \\ \hline x^1 & 0 & 8 & 3 & 14 & 10 & 12 \\ x^2 & 8 & 0 & 9 & 10 & 6 & 8 \\ x^3 & 3 & 9 & 0 & 15 & 11 & 13 \\ x^4 & 14 & 10 & 15 & 0 & 10 & 8 \\ x^5 & 10 & 6 & 11 & 10 & 0 & 8 \\ x^6 & 12 & 8 & 13 & 8 & 8 & 0 \end{array}$$

It is not hard to verify that d is indeed a distance function. Formally, before applying the neighbor-joining algorithm, we need to check that d satisfies the four-point condition. However, we will apply the neighbor-joining algorithm to d without doing such verification. We will obtain a tree $\mathcal{T}$ and compare the corresponding function $d^{\mathcal{T}}$ with d . We will see that $d^{\mathcal{T}} = d$, and hence, indeed, d satisfies the four-point condition.

We have

$$r_1 = \frac{47}{4}, r_2 = \frac{41}{4}, r_3 = \frac{51}{4},$$

$$r_4 = \frac{57}{4}, r_5 = \frac{45}{4}, r_6 = \frac{49}{4}.$$

This gives the following matrix

$$\begin{array}{c|cccccc} D & x^1 & x^2 & x^3 & x^4 & x^5 & x^6 \\ \hline x^1 & -14 & -\frac{43}{2} & -12 & -13 & -12 & \\ x^2 & & -14 & -\frac{29}{2} & -\frac{31}{2} & -\frac{29}{2} & \\ x^3 & & & -12 & -13 & -12 & \\ x^4 & & & & -\frac{31}{2} & -\frac{37}{2} & \\ x^5 & & & & & -\frac{31}{2} & \\ x^6 & & & & & & 0 \end{array}$$

In the above matrix the minimal value is $D_{13} = -43/2$. We now introduce a new OTU x^7 that will replace the pair x^1, x^3 . We place x^7 at the distance

$$d_{71} = \frac{1}{2}(d_{31} + r_1 - r_3) = 1$$

from x^1 and at the distance

$$d_{73} = \frac{1}{2}(d_{31} + r_3 - r_1) = 2$$

from x^3 , as shown in Fig. 5.6.

We will now compute distances between x^7 and each of x^2, x^4, x^5, x^6 . We have

$$d_{72} = \frac{1}{2}(d_{12} + d_{32} - d_{13}) = 7,$$

$$d_{74} = \frac{1}{2}(d_{14} + d_{34} - d_{13}) = 13,$$

$$d_{75} = \frac{1}{2}(d_{15} + d_{35} - d_{13}) = 9,$$

$$d_{76} = \frac{1}{2}(d_{16} + d_{36} - d_{13}) = 11,$$

which gives the following distance matrix for the OTUs x^2, x^4, x^5, x^6, x^7

$$\begin{array}{c|ccccc} M_d & x^2 & x^4 & x^5 & x^6 & x^7 \\ \hline x^2 & 0 & 10 & 6 & 8 & 7 \\ x^4 & 10 & 0 & 10 & 8 & 13 \\ x^5 & 6 & 10 & 0 & 8 & 9 \\ x^6 & 8 & 8 & 8 & 0 & 11 \\ x^7 & 7 & 13 & 9 & 11 & 0 \end{array}$$

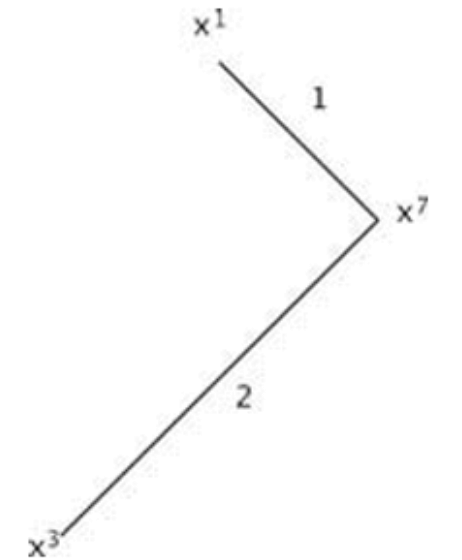


Fig. 5.6.

An Example of The Neighbor-Joining (NJ) Algorithm

For this new distance matrix we will repeat the process again and obtain

$$r_2 = \frac{31}{3}, r_4 = \frac{41}{3}, r_5 = 11,$$

$$r_6 = \frac{35}{3}, r_7 = \frac{40}{3},$$

which gives

$$D \quad x^2 \quad x^4 \quad x^5 \quad x^6 \quad x^7$$

$$x^2 \quad -14 \quad -\frac{46}{3} \quad -14 \quad -\frac{50}{3}$$

$$x^4 \quad -\frac{44}{3} \quad -\frac{52}{3} \quad -14$$

$$x^5 \quad -\frac{44}{3} \quad -\frac{46}{3}$$

$$x^6 \quad -14$$

We now introduce a new OTU x^8 that will replace the pair x^4, x^6 (note that D_{46} is minimal in the above matrix). We place x^8 at the distance 5 from x^4 and at the distance 3 from x^6 as shown in Fig. 5.7.

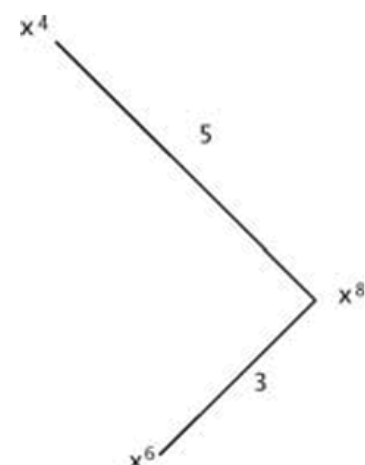


Fig. 5.7.

The distance matrix for the OTUs x^2, x^5, x^7, x^8 is

$$M_d \quad x^2 \quad x^5 \quad x^7 \quad x^8$$

$$x^2 \quad 0 \quad 6 \quad 7 \quad 5$$

$$x^5 \quad 6 \quad 0 \quad 9 \quad 5$$

$$x^7 \quad 7 \quad 9 \quad 0 \quad 8$$

$$x^8 \quad 5 \quad 5 \quad 8 \quad 0$$

On the next step of the algorithm we obtain

$$r_2 = 9, r_5 = 10, r_7 = 12, r_8 = 9,$$

and

$$D \quad x^2 \quad x^5 \quad x^7 \quad x^8$$

$$x^2 \quad -13 \quad -14 \quad -13$$

$$x^5 \quad -13 \quad -14$$

$$x^7 \quad -13$$

At this point we can group together either x^2 and x^7 , or x^5 and x^8 , since both D_{27} and D_{58} are minimal in the above matrix (the resulting tree will not depend on our choice). We group together x^5 and x^8 , that is, we introduce a new OTU x^9 , place it at the distance 3 from x^5 and at the distance 2 from x^8 as shown in Fig. 5.8, and calculate distances from x^9 to x^2 and x^7 which gives the following distance matrix for the three OTUs

$$M_d \quad x^2 \quad x^7 \quad x^9$$

$$x^2 \quad 0 \quad 7 \quad 3$$

$$x^7 \quad 7 \quad 0 \quad 6$$

$$x^9 \quad 3 \quad 6 \quad 0$$

It follows from formulas (5.2) that the above distance function is generated by the tree shown in Fig. 5.9.

Next, merging the trees from Figs. 5.6–5.9, we obtain the tree $\mathcal{T}$ shown in Fig. 5.10.

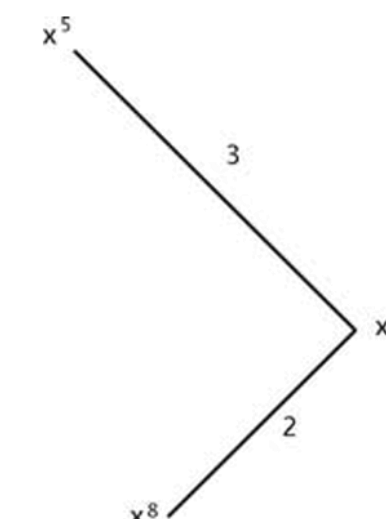


Fig. 5.8.

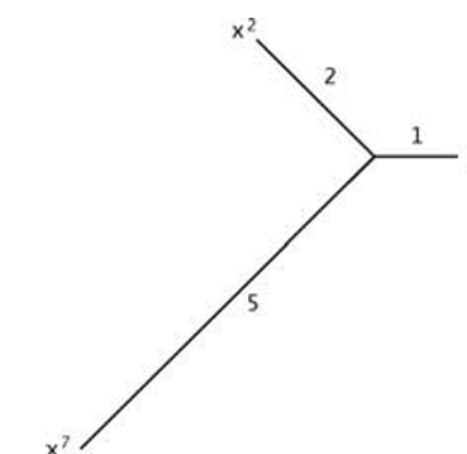


Fig. 5.9.

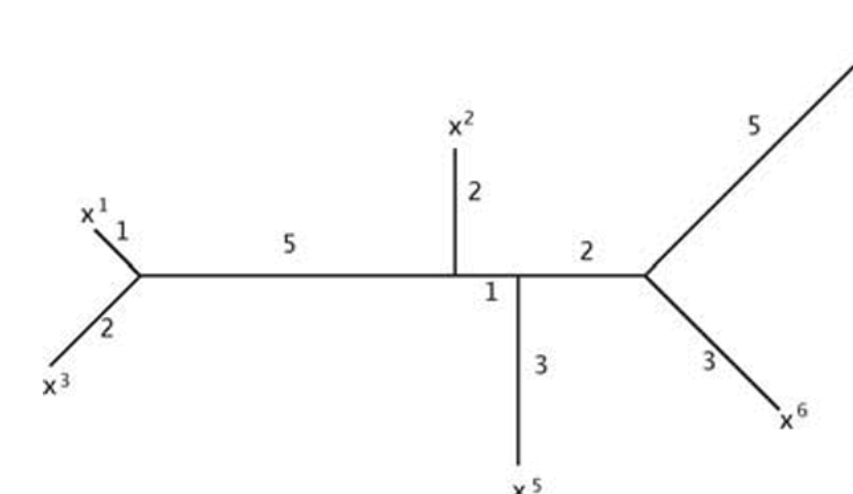


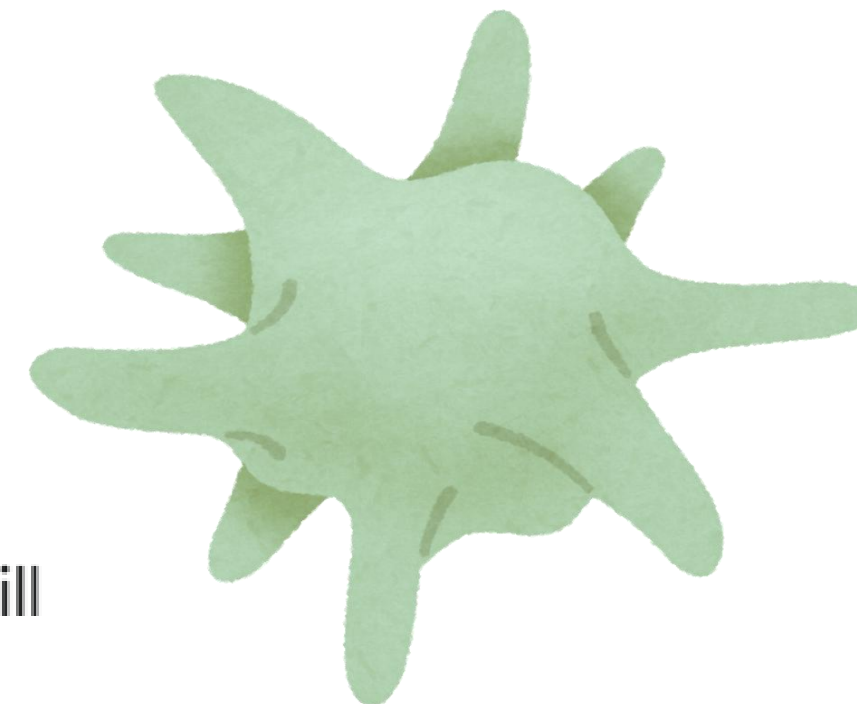
Fig. 5.10.

Distance Methods – UPGMA Method

- **UPGMA** stands for Unweighted Pair Group Method Using Arithmetic Averages.
- This is a simpler algorithm specifically for **ultrameric** distance functions.
- It assumes a **molecular clock**: evolutionary rates are constant, so branch lengths are proportional to time.
- UPGMA builds a **rooted** tree by iteratively clustering OTUs.
- It finds the two closest clusters (C_i, C_j) with the minimal distance and merges them.
- The distance between two clusters is their arithmetic average:

$$d(C_i, C_j) = \frac{1}{N(C_i)N(C_j)} \sum_{a \in C_i, b \in C_j} d_{ab}$$

- The new node (representing the merge) is placed at a height of $d(C_i, C_j)/2$.
- **Warning:** If the data is not ultrametric (i.e., does not follow a molecular clock), UPGMA will produce a tree with the **wrong topology**.



An Example of UPGMA Method

M_d	x^1	x^2	x^3	x^4	x^5
x^1	0	16	6	16	6
x^2	16	0	16	8	16
x^3	6	16	0	16	2
x^4	16	8	16	0	16
x^5	6	16	2	16	0

Example 5.8. Since d_{35} is minimal, we form a new cluster C^6 as $C^6 = \{x^3, x^5\}$. From formula (5.3) we obtain

$$d(C^1, C^6) = \frac{1}{2}(d_{13} + d_{15}) = 6,$$

$$d(C^2, C^6) = \frac{1}{2}(d_{23} + d_{25}) = 16,$$

$$d(C^4, C^6) = \frac{1}{2}(d_{43} + d_{45}) = 16.$$

We now introduce a new OTU x^6 and place it at the height $d_{35}/2 = 1$ above x^3 and x^5 , as shown in Fig. 5.20.

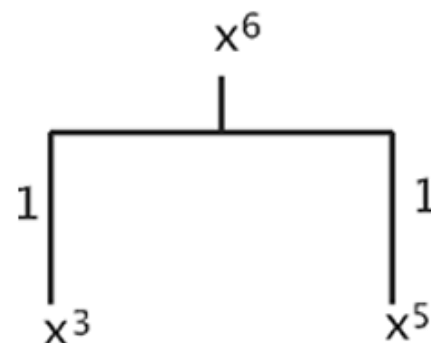


Fig. 5.20.

For the OTUs x^1, x^2, x^4 and x^6 we have the following distance matrix

M_d	x^1	x^2	x^4	x^6
x^1	0	16	16	6
x^2	16	0	8	16
x^4	16	8	0	16
x^6	6	16	16	0

Here d_{16} is minimal, and we form a new cluster C^7 as $C^7 = \{x^1, x^3, x^5\}$ (recall that x^6 is associated with the cluster $C^6 = \{x^3, x^5\}$). From formula (5.3) we obtain

$$d(C^2, C^7) = \frac{1}{3}(d_{21} + d_{23} + d_{25}) = 16,$$

$$d(C^4, C^7) = \frac{1}{3}(d_{41} + d_{43} + d_{45}) = 16.$$

We now introduce a new OTU x^7 and place it at the total height $d_{16}/2 = 3$ above x^1 and x^6 , as shown in Fig. 5.21.

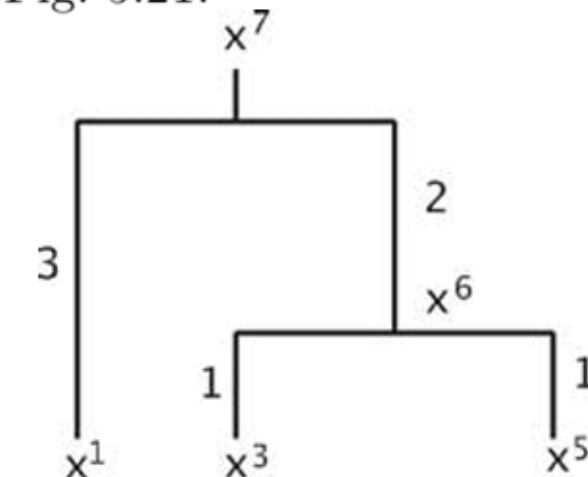


Fig. 5.21.

An Example of UPGMA Method

For the OTUs x^2 , x^4 and x^7 we have the following distance matrix

$$\begin{array}{c|ccc} M_d & x^2 & x^4 & x^7 \\ \hline x^2 & 0 & 8 & 16 \\ x^4 & 8 & 0 & 16 \\ x^7 & 16 & 16 & 0 \end{array}$$

Here d_{24} is minimal, and we form a new cluster C^8 as $C^8 = \{x^2, x^4\}$. From formula (5.3) we obtain

$$d(C^7, C^8) = \frac{1}{3 \times 2} (d_{12} + d_{32} + d_{52} + d_{14} + d_{34} + d_{54}) = 16.$$

We now introduce a new OTU x^8 and place it at the height $d_{24}/2 = 4$ above x^2 and x^4 , as shown in Fig. 5.22.

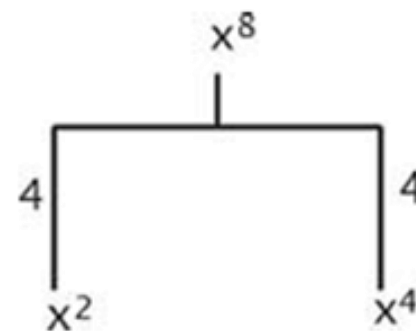


Fig. 5.22.

Finally, we place the root at the total height $d(C^7, C^8)/2 = 8$ above x^7 and x^8 which gives the tree shown Fig. 5.23.

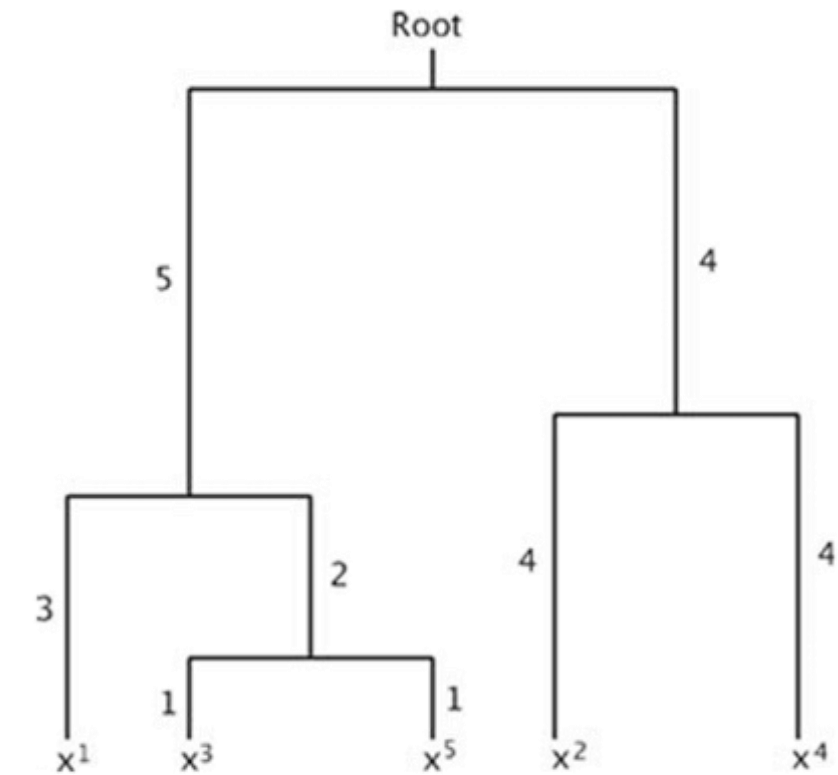


Fig. 5.23.

Maximum Likelihood Method

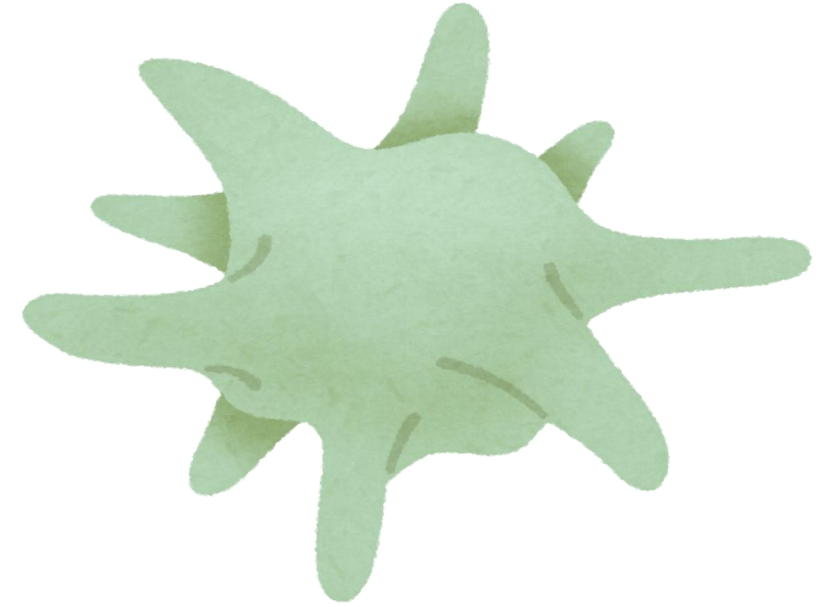
- Maximum Likelihood (ML) is a probabilistic method.
- It finds the tree (T) that maximizes the likelihood $L(D|T)$ of observing the alignment data (D).
- The likelihood $L(D|T)$ is the probability of the data given a specific tree and evolutionary model.
- Key assumptions are that sites evolve independently and follow a selected Markov model.
- The total likelihood is the product of the likelihoods for each individual site (column).

$$L(D|T) = \prod_{i=1}^n L_i(D|T)$$

- The likelihood for one site is calculated by summing probabilities over all possible ancestral states.
- Example formula (for one site, L_1) sums probabilities for all internal nodes (i,j,k).

$$L_1(D|T) = \sum_{i,j,k \in Q} \phi_i p_{ij}(u_5) p_{ik}(u_6) p_{jA}(u_1) p_{jC}(u_2) p_{kC}(u_3) p_{kG}(u_4)$$

- The Pulley Principle states that the root position does not change the likelihood due to model reversibility.
- This allows ML to search the smaller space of unrooted trees.
- ML is considered highly accurate but is computationally very slow and complex.



An Example of Maximum Likelihood Method

$\hat{x}^1 : A A G$

$\hat{x}^2 : C T C$

$\hat{x}^3 : C G G$

$\hat{x}^4 : G G A$,

and let $\mathcal{T}$ be the rooted tree shown in Fig. 5.30.

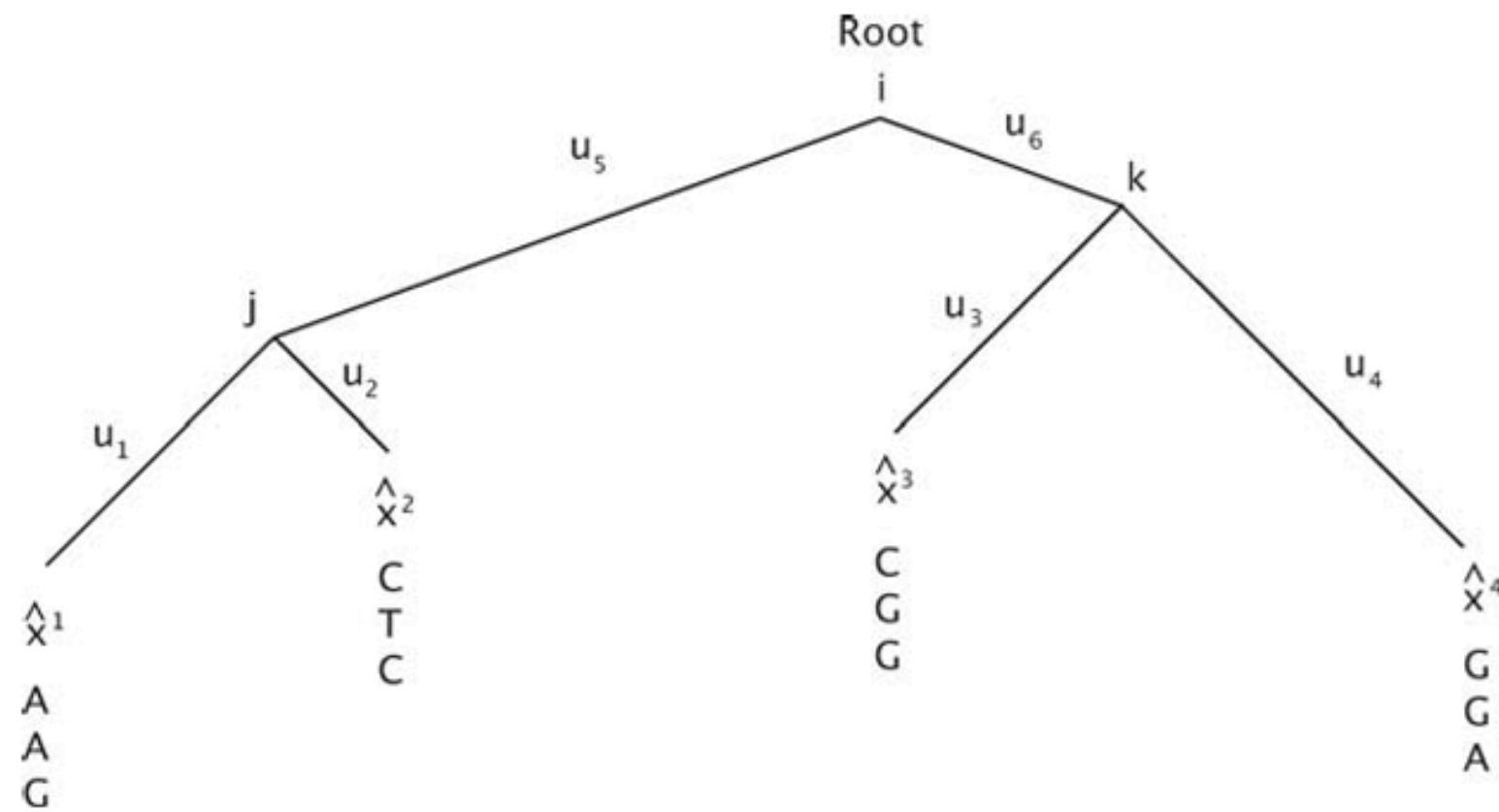


Fig. 5.30.

First of all, we define site-specific likelihoods as

$$L_1(D|\mathcal{T}) = \sum_{ijk \in \mathcal{Q}} \varphi_i p_{ij}(u_5) p_{ik}(u_6) p_{jA}(u_1) p_{jC}(u_2) p_{kC}(u_3) p_{kG}(u_4),$$

$$L_2(D|\mathcal{T}) = \sum_{ijk \in \mathcal{Q}} \varphi_i p_{ij}(u_5) p_{ik}(u_6) p_{jA}(u_1) p_{jT}(u_2) p_{kG}(u_3) p_{kG}(u_4),$$

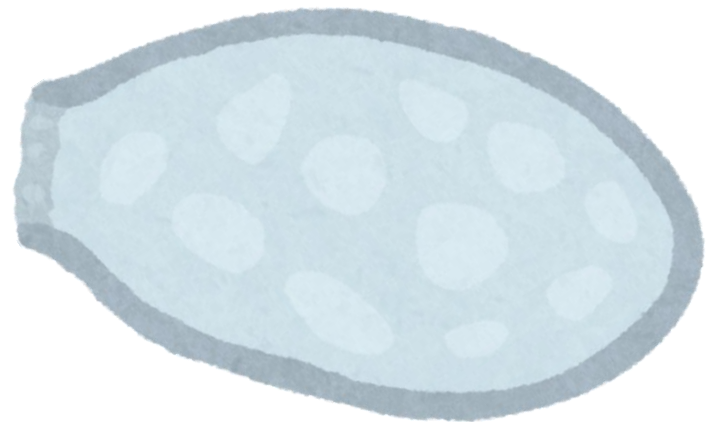
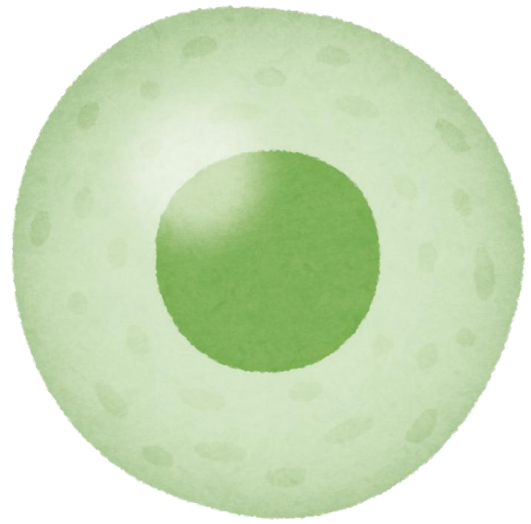
$$L_3(D|\mathcal{T}) = \sum_{ijk \in \mathcal{Q}} \varphi_i p_{ij}(u_5) p_{ik}(u_6) p_{jG}(u_1) p_{jC}(u_2) p_{kG}(u_3) p_{kA}(u_4),$$

where φ is the stationary probability distribution of the model. In the above formulas the parameter β (in the case of the Kimura and HKY models) is branch-specific. More precisely, the l th branch carries its own parameter β_l for $l = 1, \dots, 6$, with the constraint $\beta_5 = \beta_6$, which agrees with (vii) of the evolutionary assumption. For example, $p_{jA}(u_1)$ is calculated using β_1 and $p_{ij}(u_5)$, $p_{ik}(u_6)$ using $\beta_5 = \beta_6$. The *likelihood of the dataset* is then defined as the product of the site-specific likelihoods

$$L(D|\mathcal{T}) = L_1(D|\mathcal{T}) L_2(D|\mathcal{T}) L_3(D|\mathcal{T}).$$

Evolutionary Models

- Evolutionary models describe the nucleotide substitution process through time.
- Models assume that nucleotide sites evolve independently of each other.
- We use continuous-time finite Markov models.



- The states are the DNA alphabet $Q = \{A, C, G, T\}$.
- The transition probability matrix $P(t)$ gives the probability of state changes in time t .
- Models must satisfy the identity $P(t + \tau) = P(t)P(\tau)$.
- This identity implies the solution $P(t) = \exp(tQ)$.
- Q is the matrix of instantaneous change, defined as $Q = P'(0)$.
- Models have a stationary probability distribution ϕ which satisfies $\phi Q = 0$.
- Models are often time-reversible, which means $\phi_i p_{ij}(t) = \phi_j p_{ji}(t)$.

Evolutionary Models – The Jukes-Cantor Model

- The Jukes-Cantor (JC) model is the simplest substitution model.
- It has only one parameter α which is the evolutionary rate.
- It assumes all substitution probabilities are equal and base frequencies are equal (0.25).



- The matrix of instantaneous change Q is

$$Q = \begin{pmatrix} -3\alpha/4 & \alpha/4 & \alpha/4 & \alpha/4 \\ \alpha/4 & -3\alpha/4 & \alpha/4 & \alpha/4 \\ \alpha/4 & \alpha/4 & -3\alpha/4 & \alpha/4 \\ \alpha/4 & \alpha/4 & \alpha/4 & -3\alpha/4 \end{pmatrix}$$

- The resulting transition probabilities are

$$p_{ii}(t) = \frac{1}{4} + \frac{3}{4} \exp(-t\alpha) \text{ for all } i$$

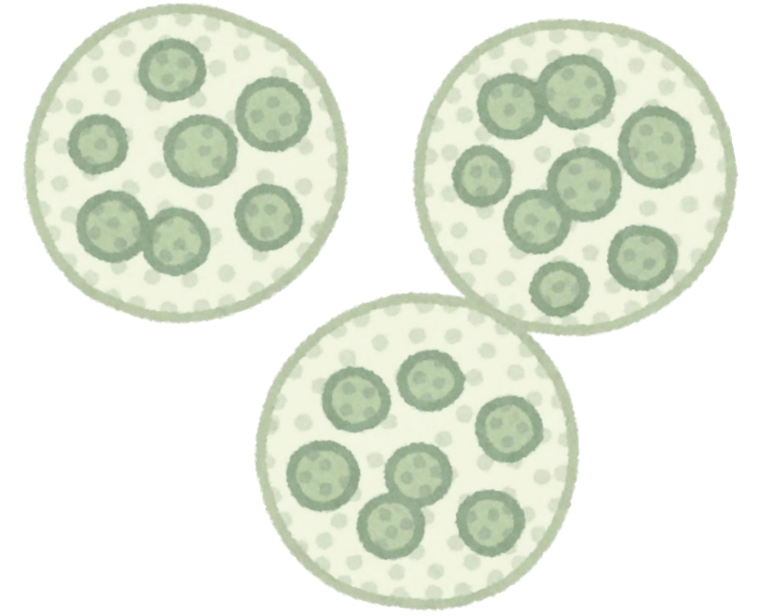
$$p_{ij}(t) = \frac{1}{4} - \frac{1}{4} \exp(-t\alpha) \text{ for all } i \neq j$$

- The stationary probability distribution is $\phi = (1/4, 1/4, 1/4, 1/4)$.
- This model is time-reversible.



Evolutionary Models – The Kimura Model

- The Kimura model (K80) is a generalization of the Jukes-Cantor model.
- It has two parameters α (evolutionary rate) and β (transition/transversion ratio).
- Transitions ($A \leftrightarrow G, C \leftrightarrow T$) are assumed to have a different rate than transversions (all others).



- The matrix of instantaneous change Q is

$$Q = \begin{pmatrix} -(2\beta + 1)\alpha/4 & \beta\alpha/4 & \alpha/4 & \beta\alpha/4 \\ \beta\alpha/4 & -(2\beta + 1)\alpha/4 & \beta\alpha/4 & \alpha/4 \\ \alpha/4 & \beta\alpha/4 & -(2\beta + 1)\alpha/4 & \beta\alpha/4 \\ \beta\alpha/4 & \alpha/4 & \beta\alpha/4 & -(2\beta + 1)\alpha/4 \end{pmatrix}$$

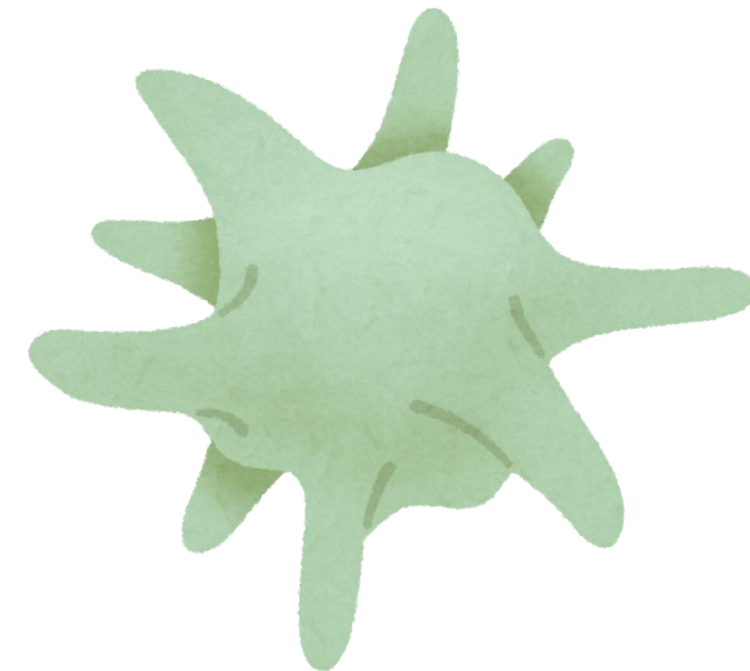
- The stationary distribution is still $\phi = (1/4, 1/4, 1/4, 1/4)$.
- This model is reversible.

- The transition probabilities $p(t)$ distinguish between transitions and transversions (formulas 5.11).

$$p_{ii}(t) = \frac{1}{4} + \frac{1}{4} \exp(-t\beta\alpha) + \frac{1}{2} \exp\left(-\frac{t(\beta + 1)\alpha}{2}\right)$$

$$p_{ij}(t) = \frac{1}{4} - \frac{1}{4} \exp(-t\beta\alpha) \text{ (for transversions)}$$

$$p_{ik}(t) = \frac{1}{4} + \frac{1}{4} \exp(-t\beta\alpha) - \frac{1}{2} \exp\left(-\frac{t(\beta + 1)\alpha}{2}\right) \text{ (for transitions)}$$



The background of the slide is a light gray, textured surface. It is decorated with various stylized, colorful microorganisms or cells. These include green circular cells with internal structures, blue star-shaped cells, and green cells with multiple small protrusions. The organisms are arranged in a border-like pattern around the central text.

THANK YOU